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Recent Advances in Interventions Targeting Remyelination and a Systematic Review of Remyelinating Effects of Approved Disease-Modifying Treatments for Multiple Sclerosis

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ABSTRACT

Background: Regenerative strategies in progressive multiple sclerosis (MS) pose a significant unmet need. Combining immunomodulatory treatment with remyelinating interventions to target the complex underlying pathogenesis appeals as the next frontier in MS therapeutic developments. Therefore, it is important to identify which disease-modifying treatments (DMT) with proremyelinating properties are most promising for future use in combination treatments. This systematic review provides an overview of preclinical and clinical research on remyelination, focusing on the effects of currently available FDA and EMA-approved DMT.

Methods: The search was conducted in accordance with the "Synthesis without meta-analysis" (SWiM) reporting guideline. The protocol was registered at PROSPERO prior to the search.

Results: Fifty-seven articles on preclinical research, three randomized controlled trials (RCTs), 29 non-randomized clinical studies, and eight reviews were included. Preclinical research suggested neuroprotective properties of various DMT. However, convincing evidence of true remyelination, either by influencing oligodendrocyte lineage cells in cell cultures or histological analysis in vivo, could only be found in studies investigating glatiramer acetate, teriflunomide, Fingolimod, Siponimod, Ponesimod, and alemtuzumab.

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Clinical trials using surrogate markers of myelin repair, such as advanced imaging and electrophysiological techniques, demonstrated promising results with glatiramer acetate, Fingolimod, Siponimod, natalizumab, alemtuzumab, and ocrelizumab. However, we found insufficient proof to claim that changes in these surrogate markers can be explained by remyelination alone.

Conclusions: Future proof-of-concept clinical trials investigating remyelinating agents in MS should consider combining outcome measures into composite endpoints. Furthermore, research efforts should be dedicated to novel biomarkers to assess repair mechanisms in MS.

1 | Introduction

Multiple sclerosis (MS) is a neuroinflammatory and neurodegenerative disease of the central nervous system (CNS), affecting more than 2.9 million people worldwide [1, 2]. Disease course is variable though traditionally classified as relapsing MS (RMS) and progressive MS (PMS) [3]. Various Food and Drug Administration (FDA) and European Medicines Agency (EMA) approved disease-modifying treatments (DMTs) effectively reduce inflammation in RMS and active PMS. Several molecules targeting underlying disease mechanisms in PMS are currently being investigated, among which Bruton's tyrosine kinase inhibitors (BTK-i) are closest to FDA and EMA registration decisions. Nevertheless, studies have resulted in conflicting results, potentially due to intra-class differences of BTK-i [4–9]. To date, no EMA or FDA-approved DMT targets disease mechanisms beyond immunomodulation [1].

1.1 | Targeting Remyelination and Neuroprotection

Experts now propose viewing MS as a disease continuum, from acute demyelination due to autoimmune inflammation leading to relapses, to neurodegeneration with chronic inflammation, lack of remyelination and axonal loss leading to progressive disability [10, 11]. These neurodegenerative processes are in part driven by failure of compensatory mechanisms, such as remyelination [12]. Remyelination indicates repair of the myelin sheath and relies on a complex interaction between various glial cell populations including oligodendrocytes (ODCs), astrocytes, and microglia [13]. Axonal function recovery does not only rely on the myelin ensheathment, but neuroprotective processes, such as limiting excitotoxicity support by neurotrophic factors and protection against oxidative stress, are important as well [14–17].

1.2 | Preclinical and Clinical Assessment of Remyelination

In vitro, myelination is mainly studied by investigating ODCs, their precursors, and the expression of specific markers of their developmental stages [18, 19]. In vivo, no single animal model mimics all disease features specific for MS [20]. The most frequently used rodent models in MS are the experimental autoimmune encephalomyelitis (EAE) model to study neuroinflammation and neurodegeneration, and the toxin-induced models such as lysophosphatidylcholine (LPC) or cuprizone (CPZ), which are often used to investigate de- and remyelination. Furthermore, local administration of ethidium bromide can cause astrocyte and ODC loss, and Theiler's murine encephalomyelitis virus (TMEV) is a viral model causing chronic progressive demyelination [20]. In addition to rodent models, translucent animals such as transgenic Xenopus tadpoles are frequently applied for developmental processes [21, 22]. The assessment of remyelination occurs through histological quantification of myelin [23], Transmission Electron Microscopy (TEM) of the sections of remyelinated tracts [24] and calculation of the inner and outer axonal diameter ratio (*g* ratio) [25].

Currently, there is a lack of validated fluid or imaging biomarkers to assess remyelination in clinical studies [19]. Surrogate markers in clinical trials investigating remyelinating interventions include advanced imaging techniques, parameters of vision (Optical Coherence Tomography (OCT) and Visual Evoked Potentials (VEP)), and clinical assessment of worsening or improvement, measured by a change in Expanded Disability Status Scale (EDSS) or a combination of clinical tests (e.g., Timed 25ft walk, 9-Hole Peg Test) [26, 27]. While VEP is used to evaluate the optic nerve in preclinical and clinical studies, restricting inclusion to cases with detectable demyelination limits its broader applicability [28–30]. Various advanced Magnetic Resonance Imaging (MRI) techniques are promising, such as Magnetization Transfer Ratio (MTR), Diffusion Tensor Imaging (DTI), Myelin Water Fraction (MWF), and q-space Myelin Map (qMM) [31]. MTR is a semiquantitative technique that measures the transfer of magnetization between free and bound proton pools in tissue [32], whereas DTI evaluates the integrity of the axons, as the diffusion of water molecules changes due to demyelination and remyelination [32]. MWF quantifies myelin water content by measuring the T2 distribution from myelin-associated water relative to total T2 [31]. QMM calculates the probability diffusion function of protons in a region of interest, estimating the presence of myelin [33, 34]. Finally, synthetic MRI uses post-processing software to visualize myelin [35] and Positron Emission Tomography (PET) allows imaging of tracers that bind specifically to components of the myelin layer [31, 32]. Most of these techniques are not specific to myelin and remain challenging due to spatial resolution, high costs, and low availability [31, 32]. Validated fluid biomarkers specific for myelin repair are not available to date [36].

1.3 | Novel and Repurposed Remyelinating and Neuroprotective Approaches in Clinical Development

Remyelination and neuroprotection depend on cross-talk between various glial cells [13], underscoring the importance of modulating various glial cells in MS to enhance remyelination and ensure neuroprotection beyond the current immunomodulatory therapy [37–40]. Although preclinical studies of novel and repurposed remyelinating interventions have yielded promising results, their translation into successful clinical trials has been met with limited

TABLE 1 | Remyelinating and neuroprotective interventions for MS in clinical development: Completed and terminated clinical trials.

Intervention	Trial design and phase	Combination with	Mode of action	Population	Outcome measures remyelination	Results	Trial registration number	References
Liothyronine	Randomized, placebo-controlled, safety, phase I	DMF, NTZ, RTX, IFNβ	Stimulating OPC differentiation, regulation of neurotrophic factors, suppression of apoptosis [42]	MS	P100 latency on VEP	Short-term safety confirmed	NCT02760056	[43]
Liothyronine	Phase 1b, single-center open label	Standard of care DMT		MS	/	Tolerability confirmed	NCT02506751	[44]
Quetiapine	Open-label, dose-finding, safety and tolerability, phase I	GA, IFNβ, DMF, FTY, no DMT	Stimulating proliferation and maturation of ODC, release of neurotrophic factors, inhibiting activated microglia and astrocytes [45]	RRMS and PMS	/	Intolerable	NCT02087631	[46] (Abstract only)
Transvaginal estriol	Open label	Standard of care DMT	Promotion OPC differentiation [47]	RRMS	VEP, OCT	Improvement VEP and OCT	NCT03774407	[48] (Abstract only)
Bazedoxifene	Randomized, placebo-controlled, phase II	Standard of care DMT	Promotion OPC differentiation [47]	RRMS	MWF, VEP	Not yet reported	NCT04002934	[49] (Protocol)
ACTH	Randomized, controlled	Exclusion of NTZ, RTX, CYC	Promotion differentiation OPC and protection ODC from oxidative and inflammatory damage [50]	RRMS, SPMS, active disease	MWF, EDSS	Terminated due to slow enrollment	NCT02446886	Only posted on Clinicaltrials.gov
Opicinumab/ BIIB033	Randomized, blinded, placebo-controlled, serial-cohort, multiple ascending dose, phase I	IFNβ, GA, no DMT	Human monoclonal antibody against LINGO-1 (inhibitor of ODC differentiation)	RRMS, SPMS	MTR	Safety confirmed	NCT01244139	[51]

(Continues)

TABLE 1 | (Continued)

Intervention	Trial design and phase	Combination with	Mode of action	Population	Outcome measures remyelination	Results	Trial registration number	References
Opicinumab	Randomized, placebo-controlled, dose-ranging, phase II	IFNβ		RRMS and SPMS with active disease activity	Disability change (EDSS, 9HPT, PASAT)	No dose–response in confirmed disability improvement (composite outcome EDSS, T25FW, 9HPT, PASAT)	NCT01864148	[52]
Opicinumab	Randomized, placebo-controlled, phase II	DMF, NTZ, IFNβ		RMS	Multicomponent clinical score (EDSS, 9HPT, T25FW)	Terminated since part 1 did not meet primary nor secondary endpoints	NCT03222973	[53] (Abstract only)
Oral BIIB061	Randomized, placebo-controlled, dose-ranging, phase II	IFNβ, GA		RMS, SPMS	Overall disability response score (EDSS, T25FW, 9HPT), MTR, DTI	Withdrawn due to the lack of stronger preclinical effects of BIIB061 on remyelination relative to opicinumab	NCT04079088	Only posted on Clinicaltrials.gov
Mesenchymal stem cells	Randomized, cross-over phase I/II trial		Suppression autoimmune response and activation OPC [54]	RMS, some sites: SPMS, PPMS	Imaging, EDSS, MSFC, OCT	Acceptable safety, but no effect on MRI marker of acute inflammation Toulouse site: Terminated due to default inclusion	Merge of partially independent clinical trials NCT01745783NCT02035514 NCT02403947 NCT02239393 NCT01854957 NCT01606215 EudraCT, 2012-000518-13 EudraCT, 2015-000137-78	[55]
Mesenchymal stem cells	Phase I–II	/		MS	Change in EDSS	Acceptable safety	NCT00781872	[56]
Intrathecally transplanted human fetal NPCs	Open-label, dose-finding, phase I	/	Trophic support, immunomodulation	PMS	/	Safety outcome reached	NCT03269071	[57]

(Continues)

TABLE 1 | (Continued)

Intervention	Trial design and phase	Combination with	Mode of action	Population	Outcome measures remyelination	Results	Trial registration number	References
Domperidone	Randomized, controlled, phase II	GA, IFNβ, DMF, TF, FTY	Increase prolactin level [58]	RRMS with enhancing lesions	DTI, MTI	Recruitment target was not met	NCT02493049	[59]
Nanocrystalline gold (CNM-Au8)	Randomized, placebo-controlled, phase II	Standard of care DMT	Oral liquid suspension of gold nanocrystals Promotion OPC differentiation and ODC maturation, antioxidant [60]	RRMS with chronic optic neuropathy	LCLA, VEP, OCT, MTR, MWF, MSFC	Terminated early due to COVID-19-related enrollment challenges Improvement in LCLA and SDMT in double-blind period	NCT03536559	[61] (Abstract only)
Nanocrystalline gold (CNM-Au8)	Open-label, long-term extension study	Standard of care DMT		RRMS with chronic optic neuropathy	LCLA, VEP, OCT, MTR, MWF, MSFC	Improved neurological function	NCT04626921	[62]
Clemastine	Randomized, placebo-controlled, cross-over, phase II	Standard of care DMT	M1 muscarinic receptor antagonist Promotion OPC differentiation Reduction inflammation [63]	Stable RRMS	VEP, MTR, MWF	Primary endpoint met, shortening of P100 latency delay with a reduction of 1.7 ms/eye Increased MWF	NCT02040298	[64, 65]
Bexarotene	Randomized, placebo-controlled, phase II	DMF	Retinoid Xreceptor agonist Promotion of OPC differentiation [66]	RRMS	VEP, MTR	Poor tolerability and negative primary outcome	ISRCTN14265371	[67]
Phenytoin	Randomized, placebo-controlled, phase II	NTZ excluded	Blockade voltage-gated sodium channels Proliferation and differentiation OPC [68]	Acute optic neuritis; Exclusion if RRMS duration > 10 years	OCT, VEP, DTI	Reduction in the extent of RNFL loss with phenytoin compared with placebo	NCT01451593	[69]

(Continues)

TABLE 1 | (Continued)

Intervention	Trial design and phase	Combination with	Mode of action	Population	Outcome measures remyelination	Results	Trial registration number	References
GSK239512	Randomized, placebo-controlled, phase II	IFNβ, GA	Histamine receptor antagonist/ inverse agonist Promotion OPC differentiation	RRMS	MTR, EDSS worsening	A small but positive effect of was observed using lesional MTR	NCT01772199	[70]
Olesoxime	Randomized, placebo-controlled, phase Ib	IFNβ	Cholesterol-like compound Promotion ODC maturation [71]	RRMS	DTI, MTR	Safe, no difference in exploratory MRI endpoints	NCT01808885	[72] (Abstract only)
EHP-101	Open-label, dose-finding, phase IIa		Oral lipidic cannabinoid, ODC differentiation, prevention microglial activation and astrogliosis [73]	RMS, active SPMS	MSFC	Temporary recruitment pauses to re-assess the protocol design, specifically the eligibility criteria; no update since 21 oct 2022	NCT04909502	Only posted on Clinicaltrials.gov
rTMS	Randomized, sham-controlled phase I clinical trial			RMS, SPMS	DTI, MTR	Safe, well-tolerated, no changes in MRI, cognition or motor performance	ACTRN12619001196134	[74]

Abbreviations: 9HPT, 9-hole peg test; DTI, Diffusion Tensor Imaging; EDSS, Expanded Disability Status Scale; LCVA, high/low contrast visual acuity; MSFC, Multiple Sclerosis Functional Composite; MTI, Magnetization Transfer Imaging; MTR, Magnetization Transfer Ratio; MWF, Myelin Water Fraction; OCT, Optical Coherence Tomography; PASAT, Paced Auditory Serial Addition Test; SDMT, symbol digit modalities test; T25FW, Timed-25 ft Walk; VEP, Visual Evoked Potential.

TABLE 2 | Remyelinating and neuroprotective interventions for MS in clinical development: Starting and ongoing clinical trials.

Intervention	Clinical trial design and phase	Combination with	Proposed mode of action	Glial cell types targeted	Study population	Primary outcome measure	Other outcome measures focused on remyelination and neuroprotection	Trial registration number	References
Clemastine	Randomized, placebo-controlled, phase I–II	Standard of care DMT or no DMT	M1 muscarinic receptor antagonist	OPC	RRMS	MRI corpus callosum MWF, T1 relaxation time, UTE Fraction	Optic radiation, corticospinal tract, lesion of interest, whole brain MWF, T1 relaxation time, UTE fraction	NCT05359653	[64]
Clemastine	Phase III	Standard of care DMT or no DMT	M1 muscarinic receptor antagonist	OPC	MS	Versional Dysconjugacy Index (VDI) measured by infrared oculography	Other infrared oculography parameters, SDMT, EDSS, HCVA and LCVA. PRO: quality of life, visual complaints, fatigue	NCT05338450	[75]
Metformin and Clemastine	Phase II	Standard of care DMT	• Antioxidant, modulating microglia toward M2 phenotype, AMPK activation • M1 muscarinic receptor antagonist	OPC, microglia	RRMS	P100 latency of the full-field VEP	Multifocal VEP, lesional MTR, other parameters of vision	NCT05131828	[76]
Repurposed drugs (metformin and alpha-lipoic acid)	Phase II/III	Standard of care DMT or no DMT	• Antioxidant, modulating microglia toward M2 phenotype, AMPK activation • Antioxidant	OPC, microglia	PPMS and SPMS	Whole-brain atrophy rate, time to disability progression	Imaging: atrophy, T2 lesion quantification, time to disability progression, EDSS, T25FW, SDMT, 9HPT, LCVA, relapse rate, MSIS19, MSWS, quality of life Fatigue, pain assessment	ISRCTN14048364	[77]
Metformin	Phase I	Standard of care DMT	Antioxidant, modulating microglia toward M2 phenotype, AMPK activation	OPC, microglia	PPMS and SPMS	Safety, new T2 lesions	Imaging: cortical thinning, thalamic atrophy, Rim lesions SDMT, CVLT, PACC, PASAT, plasma Nfl	NCT05349474	

(Continues)

TABLE 2 | (Continued)

Intervention	Clinical trial design and phase	Combination with	Proposed mode of action	Glial cell types targeted	Study population	Primary outcome measure	Other outcome measures focused on remyelination and neuroprotection	Trial registration number	References
Metformin	Phase II	Standard of care DMT or no DMT	Antioxidant, modulating microglia toward M2 phenotype, AMPK activation	OPC, microglia	MS	Safety, percent change N-acetyl aspartate cortex	Conventional MRI, MWF, EDSS, SDMT, T25FW, 9HPT, BVMT, CVLT	NCT06463743	
Metformin	Randomized, placebo-controlled, phase II	Standard of care DMT or no DMT	Antioxidant, modulating microglia toward M2 phenotype, AMPK activation	OPC, microglia	PPMS and SPMS	T25FW	SDMT, 9HPT, EDSS WBV, T2LV, T1LV, DTI, PRL	NCT05893225	[78]
Testosterone	Phase II	NTZ, FTY, PON, OFA, OCR	Neural androgen receptors Neuroprotection	OPC	MS	Imaging: thalamic atrophy, transverse diffuse lesions	T1 lesions, T2 lesions, FLAIR lesions, MTI, BICAMS, quality of life, fatigue, HADS, EDSS, safety	NCT03910738	[79]
Ifenprodil	Phase II	Standard of care DMT or no DMT	GluN2B-specific NMDAR antagonist	OPC	RRMS	P100 latency VEP	Imaging: atrophy rate, PET-MR, MTR, VEP, OCT, serumNfl, safety	NCT06330077	[80]
PIPE-307	Phase II	Stable on a single prior standard of care DMT	Selective antagonist of the M1 muscarinic receptor	OPC	RRMS	Safety, LCLA	LCLA, T25FW, 9HPT, serumNfl, blood concentration PIPE-307, imaging of myelination	NCT06083753	[81]
PIPE-791	Phase I		LPA1 antagonist	OPC	PPMS or SPMS	LPA1 occupancy on brain and lung scan	Safety, pharmacokinetics	NCT06683612	[82]
MIND diet	Phase II	Standard of care DMT or no DMT	Reduction of oxidative stress and inflammation, gut-brain axis, stimulate secretion of neurotrophic factors		MS	PlasmaNfl	NFI-MS, EDSS, SDMT, CVLT, BVMT-R, 9HPT, T25FW, 2MWT, MSIS-29, MSCS, MIND diet score, ISIGLTEQ, GFAP level, NHANES, LTL	NCT06992115	

(Continues)

TABLE 2 | (Continued)

Intervention	Clinical trial design and phase	Combination with	Proposed mode of action	Glial cell types targeted	Study population	Primary outcome measure	Other outcome measures focused on remyelination and neuroprotection	Trial registration number	References
N-acetyl Cysteine	Phase II	Standard of care DMT or no DMT	Glutathione (GSH) precursor with antioxidant properties, ODC survival	ODC	PMS	Safety, tolerability, imaging: atrophy	9HPT, T25FW, SDMT Imaging metrics and changes multi-sensor device	NCT05122559	[83]
Closed loop trans-auricular vagus nerve stimulation system with motor task	Randomized, sham-controlled, trial	Standard of care DMT or no DMT	Stimulation of the vagus nerve	Microglia Astrocytes	RRMS, RMS, SPMS	9HPT, EDSS, T25FW, composite endpoint, Fugl-Meyer assessment	—	NCT06641271	
Aerobic exercise	Phase I–II				MS	SSEP	6MWT, TUG, T25FW, 9HPT, MWF, SDMT, MFIS, quality of life, IPAQ-SF, aerobic fitness exercise, heart rate	NCT04539002	[84]
SetPoint system (vagus nerve stimulation)	Randomized, sham-controlled trial	Standard of care DMT	Stimulation of the vagus nerve		RRMS	Safety	VEP latency, imaging: T2 lesions, PRL, MTR, DTILCLA/HCLA, visual function EDSS, MSFC, MFIS	NCT06796504	[85]
Repetitive TMS	Randomized, sham-controlled, phase II	Standard of care DMT or no DMT	Transcranial magnetic stimulation	ODC	MS	MS functional composite score	Safety, HADS, quality of life, Fatigue, sleep quality, imaging	ACTRN12622000064707	[86]
Allogeneic Umbilical Cord Mesenchymal Stem Cells	Phase I/IIa	Failure on standard of care DMT			RRMS, SPMS	Safety, EDSS	MRI parameters, T25FW, 9HPT, SDMT, PASAT, MSFC, quality of life, OCT, relapse rate, MSFC	NCT05532943	[87]

Abbreviations: 2MTWT, 2-minute walking test; 6MWT, 6-minute walking test; 9HPT, 9-hole peg test; BAI, Biological Aging Index; BICAMS, Brief International Cognitive Assessment for Multiple Sclerosis; BVMT, Brief Visuospatial Memory Test; CVLT, California Verbal Learning Test; GFAP, Glial fibrillary acidic protein; GLTEQ, Godin Leisure-Time Exercise Questionnaire; HADS, Hospital Anxiety and Depression Score; HCVA/LCVA, High/low contrast visual acuity; IPAQ-SF, International Physical Activity Questionnaire Short Form; ISI, Insomnia Severity Index; MIND, Mediterranean-DASH Intervention for Neurodegenerative Delay; MSCS, Multiple Sclerosis Cognitive Scale; MSWS, Multiple Sclerosis Walking Scale; MTI, Magnetization Transfer Imaging; MWF, Myelin Water Fraction; NFI-MS, Neurological Fatigue Index-MS; Nfl, neurofilament light chain; ODC, oligodendrocyte; PACC, Preclinical Alzheimer's Cognitive Composite; PASAT, Paced Auditory Serial Addition Test; PRL, paramagnetic rim lesions; PRO, patient reported outcomes; SDMT, Symbol Digit Modalities Test; SSEP, Somatosensory Evoked Potentials; T25FW, Timed-25ft Walk; TUG, Timed Up and Go; UTE, Ultra Short Echo Time; VDI, Versional Dysconjugacy Index; VEP, Visual Evoked Potential.

success [41]. Tables 1 and 2 show completed and ongoing trials of novel and repurposed remyelinating approaches in MS.

1.4 | Aim of the Review

Combination treatment of immunomodulatory therapy and remyelinating interventions should be the path forward to optimize outcomes in MS patients [88]. Therefore, identifying which of the approved DMTs pair best with novel or repurposed agents is highly relevant [89, 90]. Given MS affects people worldwide and the cost of remyelination and neuroprotective therapies—particularly in combination with immunomodulatory drugs—will be a key barrier to global access, currently approved DMTs remain an important foundation for add-on treatments. Besides, the role of the adaptive immune system in remyelination is emerging, challenging the general view that approved DMTs have no relevant effect on remyelination [91, 92]. Therefore, this review aims to provide an up-to-date overview of the remyelinating effects of currently available FDA- and EMA-approved DMTs, used alone or in combination, in in vitro and in vivo models as well as in MS patients.

2 | Methods

This systematic review was performed according to the reporting guideline "Synthesis without meta-analysis" (SWiM) [93] and registered in the International Prospective Register of Systematic Reviews (PROSPERO) prior to the search (last search in July 2025). Eligibility criteria were agreed upon by five researchers experienced in MS-related research (MD, MC, JD, TR, BW). Keywords were "remyelination", "multiple sclerosis," and a list of approved DMT. The following databases were screened: MEDLINE (via PubMed), Cochrane Library, Web of Science, and Turning Research into Practice (TRIP). Ongoing registered trials were included using Clinicaltrials.gov and the EU clinical trial register. Snowball sampling was performed to identify relevant reports not found through the search strategy. The literature search and initial screening on title/abstract/keywords were conducted by a single reviewer (A.V.d.K.), in consultation with the co-authors. Original research articles, systematic reviews, reviews, and clinical trial reports (phase I, II, III) were included. Case reports, books, conference or congress abstracts, and expert opinions were excluded. Only English articles were included, published in the last 15 years. Ongoing studies for which the last posted information dated more than 2 years ago and without a publication since were excluded. Screening on full text was performed independently by two reviewers (A.V.d.K. and E.V.). Only preclinical and clinical research focusing on the remyelinating properties of current FDA- and EMA-approved DMT were included. Conformable to the protocol, this review additionally incorporated articles with indirect measurements of remyelination, including neuroprotection. Full search strategy and search string are described in the Supporting Information.

Data collection was performed by one reviewer (A.V.d.K.) and included DOI, NCT number, title, publication date, study design, sample size, outcome measures, duration, follow-up, and presence of a control group (Supporting Information). Studies that passed the screening process were imported into Endnote version 20.4.1. Risk of bias was assessed by one reviewer (A.V.d.K.), using Cochrane's Risk of Bias (RoB2) tool [94] and ROBINS-I tool [95] for RCTs and non-randomized controlled and uncontrolled trials, respectively. For preclinical trials, the SYRCLE RoB tool [96] (in vivo studies) and the QUIN risk of bias tool [97] (in vitro studies) were used.

3 | Results

The systematic search yielded a total of 465 records. After screening for relevance to the subject based on title, abstract, and full text, a total of 97 manuscripts were included in the synthesis. A PRISMA flowchart (Figure 1) depicts the process. The search resulted in three RCTs, 29 non-randomized human clinical trials, eight reviews, and 57 preclinical research articles. Data were not suitable for subgroup analysis. Information on application, mode of action, and possible remyelinating effects is depicted in Table 3 and Figure 2.

3.1 | Platform Treatment: Injectables

3.1.1 | Interferon-Beta (IFNβ)

Four studies described the effects of IFNβ on remyelination and neuroprotection in a preclinical setting. IFNβ did not promote oligodendrocyte precursor cell (OPC) differentiation under basal or inflammatory conditions in mouse OPCs [166] and had no significant effect on cell proliferation, neuronal, and ODC lineage cell differentiation in mouse neural progenitor cells (NPCs), but the reduction of apoptosis suggested a neuroprotective effect [167]. One study demonstrated a dose-dependent effect on neural stem cell fate in human NPCs [104], whereas low concentrations of IFNβ resulted in differentiation into astrocytes, while high concentrations resulted in an increased amount of OPC differentiation [104]. One study demonstrated a synergistic effect of combination treatment with dimethyl fumarate (DMF) and IFNβ in an EAE model leading to less demyelination compared to monotherapy with either drug [168].

Fourteen studies reported clinical research. One retrospective cohort found no association between treatment and reduction in disability progression [169], but another retrospective analysis reported reduced gray matter atrophy rates after IFNβ-1a treatment and established an association between disability progression and changes in gray matter fraction [170]. In studies focusing on optic neuritis, two independent investigations using OCT, visual fields, and VEP failed to demonstrate a significant effect of IFNβ [171, 172]. Meanwhile, an investigation of the effects of IFNβ-1a in voxel-wise (VW) MTR revealed a greater change in the volume of brain tissue with decreasing MTR, indicating demyelination. Despite this, there was a notable change in normal appearing brain tissue (NABT) volume with increasing VW-MTR in the IFNβ group compared to healthy controls (HCs) over a 12-week period [173]. Comparative MTR studies revealed that IFNβ treatment resulted in a larger volume of demyelination (decreasing VW-MTR) compared to the natalizumab (NTZ)-treated group and HCs [174] and more unrepaired

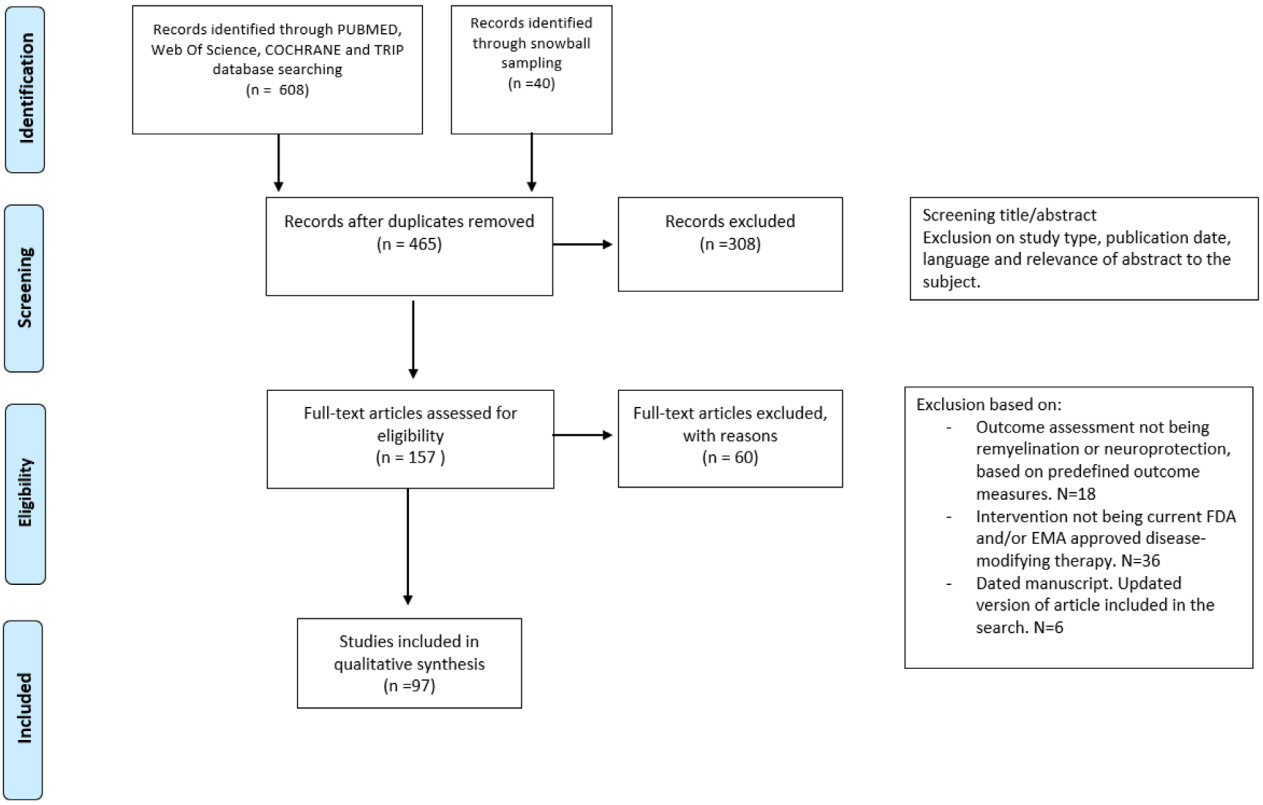

FIGURE 1 | PRISMA flowchart.

lesional damage compared to glatiramer acetate (GA) in lesional fat-suppression MTR analysis [175].

3.1.2 | Glatiramer Acetate (GA)

Seven preclinical studies have reported neuroprotective effects of GA. These studies highlight GA's ability to induce T cells that enhance the expression of several neuroprotective factors crucial for neuronal function and survival, such as brain-derived neurotrophic factor (BDNF), neurotrophin-3 and -4, as well as insulin-like growth factor-1 and -2. These findings suggest that GA may play a role in supporting neuronal health through the modulation of proteins involved in neuroprotection [107–109, 176, 177]. Moreover, one study described reduced axonal damage after GA treatment [178]. In various rodent models (EAE, LPC, CPZ), treatment with GA resulted in a suppression or reduction of the severity of the clinical phenotype [111, 179] and improved behavioral testing [111, 177], suggesting a functional effect. Histological analysis in animal models demonstrated less demyelination and enhanced remyelination compared to untreated animals, which was evidenced by an increased number of OPCs [109], elevated markers of differentiation and proliferation of ODC lineage cells [112], increased staining of myelin markers, and restored myelin deposition and structure [110]. However, in the LPC model, no significant differences in mean *g* ratio compared to controls were demonstrated [110].

Six studies investigated clinical and radiological outcomes. Analysis of T1 lesion evolution and acute black holes found that, in GA-treated patients, lesions did not become chronic but stayed or turned iso-intense. Since T1 hypo-intensities on MRI reflect axonal damage linked to disability, these findings are suggestive of increased repair [180]. Lesional fat-suppression MTR showed less unrepaired damage in GA-treated patients compared to IFNβ-treated groups [175]. A single open-label study in GA-treated patients showed diffusion restriction recovery, suggesting repair [66]. MRI spectroscopy revealed a higher N-acetyl aspartate to creatinine ratio, indicating better neuronal and axonal metabolic health [181].

3.2 | Oral Treatment

3.2.1 | Teriflunomide (TF)

Five preclinical studies with TF investigated remyelination. In vitro studies established an effect of TF on proliferation and differentiation in glial cultures, OPC cultures, and myelinating neuron/glia co-cultures [116, 117]. One study demonstrated that only short stimulation (24 h) with TF led to OPC differentiation, generation of myelinating ODCs, and upregulation of genes involved in myelination [116]. In the CPZ model, both constant and pulsed TF administration improved spontaneous remyelination in the caudal corpus callosum, as shown by TEM and increased myelin gene expression [115]. Furthermore, in both the tadpole and LPC models, TF administration resulted in enhanced remyelination, as demonstrated by immunolabeling of the optic nerve and TEM of the lesion area, respectively [117]. One in vivo study using the TMEV model suggested TF reduces excitotoxicity by demonstrating

TABLE 3 | Application, anti-inflammatory mode of action and remyelinating effects of current FDA/EMA-approved DMT.

DMT	Application	Anti-inflammatory mode of action	Remyelinating effects (preclinical) and possible mode of action
Interferon-beta (IFNβ)	Recombinant forms of IFNβ (IFN beta-1a, IFN beta-1b, peg-IFN beta-1b) administered by subcutaneous (SC) or intramuscular (IM) injection. Platform treatment in RMS and secondary PMS with active disease [98–101]	Downregulation of Major Histocompatibility Complex class II expression on antigen-presenting cells and induces T-cell production of interleukin 10, resulting in an increased expression of anti-inflammatory factors while downregulating the expression of pro-inflammatory cytokines [102, 103]	+/− Dose-dependent effect on neural stem cell fate (in one preclinical study [104])
Glatiramer acetate (GA)	Synthetic copolymer whose composition resembles myelin basic protein (MBP). Platform treatment by SC injection of 20 mg daily or 40 mg three times weekly [105]	Shifts the T-cell response from a pro-inflammatory to an anti-inflammatory pathway and influences B cells to favor the production of anti-inflammatory cytokines. GA-specific T cells are able to cross the blood–brain barrier and secrete in situ anti-inflammatory cytokines [106, 107]	+ GA-induced T cells enhance expression of BDNF, neurotrophin-3, neurotrophin-4, insulin-like growth factor-1 and insulin-like growth factor-2 (in five preclinical studies [108–112])
Teriflunomide (TF)	Oral treatment for RMS taken once daily [113]	Cytostatic effect on T- and B-cell proliferation due to inhibition of dihydroorotate dehydrogenase, a mitochondrial enzyme involved in pyrimidine biosynthesis and expressed in proliferating lymphocytes [114]	+ Unknown mode of action (three preclinical trials [115–117])
Fumarates	Dimethyl fumarate (DMF) and diroximel fumarate (DRF) are oral treatment for RMS with an intake twice daily [118, 119]	The active metabolite monomethylfumarate (MMF) activates nuclear factor erythroid-derived 2 (Nrf2) pathway, resulting in a change in immune cell composition and phenotype and reduction in CNS [120–122]	−
Fingolimod (FTY)	Oral treatment taken for patients with active RMS [123]. FTY has a specificity for S1P1, 3 and -5 receptors	S1PR modulator that prevents egression of lymphocytes from lymph nodes into the blood [124]	+ Dose- and duration dependent effect on NPC directly and indirectly via p38MAPK, ERK1/2, and CREB pathways (> 15 preclinical trials, however conflicting results [125–139])
Siponimod (SPN)	Oral treatment taken once daily. In some countries it is indicated for RMS and secondary PMS, while in others it is only used for active SPMS [140, 141]. SPN binds to S1P1 and S1P5 receptor	Selective S1P1 and S1P5 receptor modulator [141–142] that prevents egression of lymphocytes from lymph nodes into the blood	+ Effect through S1P5 receptor modulation (four preclinical trials [21, 143–145])

(Continues)

TABLE 3 | (Continued)

DMT	Application	Anti-inflammatory mode of action	Remyelinating effects (preclinical) and possible mode of action
Ozanimod (OZN)	Oral treatment taken once daily for active RMS (EMA) and active SPMS (FDA) [146, 147]. OZN binds to S1P1 and S1P5 receptors	Selective S1P1 and S1P5 modulator [148] that prevents egression of lymphocytes from lymph nodes into the blood	–
Ponesimod (PON)	Oral treatment taken once daily for active RMS (EMA) and active SPMS (FDA) [149, 150]. PON is specific for S1P1 receptors	Selective S1P1 receptor modulator [148] that prevents egression of lymphocytes from lymph nodes into the blood	+/– Unknown mode of action (one preclinical trial [151])
Natalizumab (NTZ)	High efficacy treatment administered as an injection (IV or SC) every 4 weeks in patients with active RMS [152]	Humanized monoclonal antibody targeted against four alfa-integrin molecules on leukocytes, blocking the transmigration of immune cells to the CNS [153]	–
Ocrelizumab (OCR)	High efficacy treatment for active RMS or early PMS, IV injection every 6 months [154]	Recombinant humanized anti-CD20 antibody that selectively depletes circulating B cells, but spares CD20 negative plasma cells [155]	–
Ofatumumab (OFA)	Treatment for active MS, SC injection every month [156]	Fully human monoclonal anti-CD20 antibody that selectively depletes circulating B cells [157]	–
Cladribine (CLAD)	Immune reconstitution therapy administered in yearly treatment courses for 2 years. It has been approved for patients with highly active RMS [158]	Purine analogue which causes a selective interference with DNA synthesis in activated T and B cells, resulting in a targeted lymphocyte cell death and repopulation of these subsets [159, 160]	–
Alemtuzumab (ALEM)	Immune reconstitution therapy administered IV for two initial treatment courses, with up to twoadditional treatment courses if needed. It is indicated for highly active RMS [161]	Humanized anti-CD52 monoclonal antibody, targeted against a surface glycoprotein predominately found on differentiated lymphocytes, which results in a depletion and repopulation of lymphocytes, leading to long-lasting changes in adaptive immunity [162, 163]	+ Increased neurotrophic factors (BDNF, platelet-derived growth factor, ciliary neurotrophic factor) (two preclinical trials [164, 165])

Note: + Evidence +/– questionable – No Evidence.

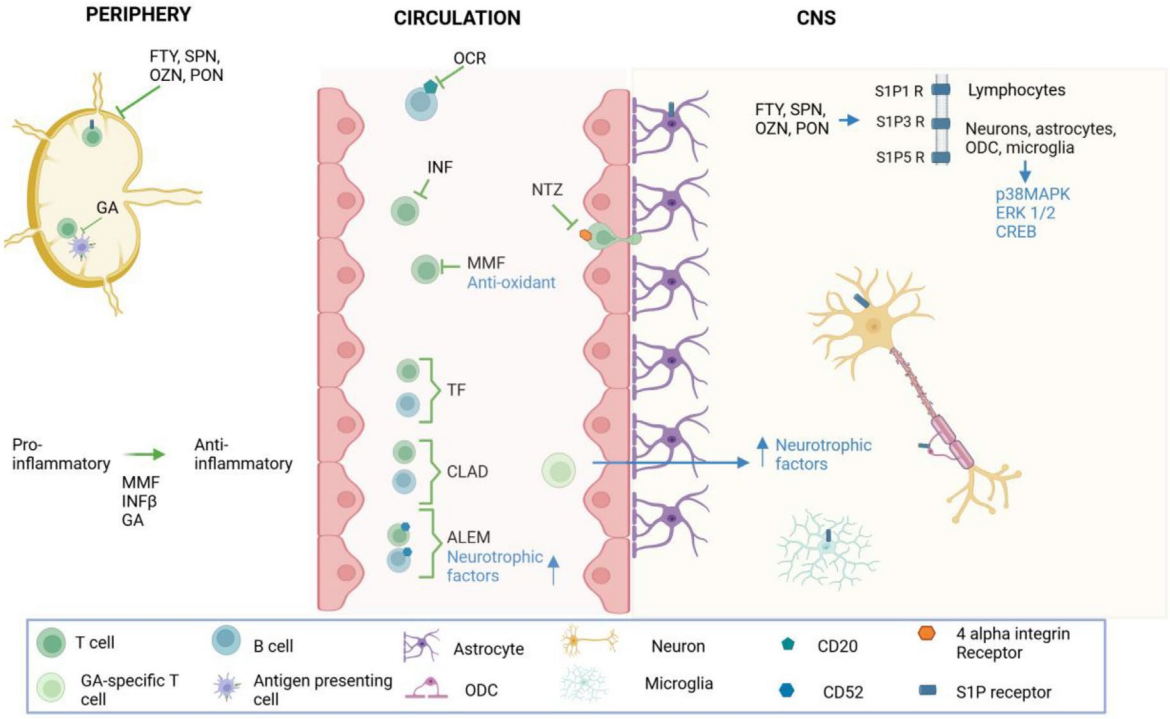

FIGURE 2 | Anti-inflammatory mode of action (green) and possible effect on remyelination (blue) of current disease-modifying therapies. ALEM, alemtuzumab; CLAD, Cladribine; FTY, Fingolimod; GA, glatiramer acetate; INFβ, interferon beta; MMF, monomethyl fumarate; NTZ, natalizumab; OCR, ocrelizumab; OZN, ozanimod; PON, Ponesimod; SPN, Siponimod; TF, teriflunomide. *Image created* via *Biorender.*

a decreasing trend of glutamate excess on MR spectroscopy after treatment [182].

Two clinical studies focused on remyelination and neuroprotection. A retrospective analysis of the TEMSO trial found that patients treated with TF experienced less brain volume loss compared to untreated patients [183]. Additionally, the study linked greater brain volume loss to a higher risk of disability progression, suggesting that TF may help mitigate brain atrophy and reduce the likelihood of worsening disability [183]. A prospective comparative study in patients with acute optic neuritis demonstrated a significant amelioration of VEP and low contrast visual acuity compared to GA- or IFNβ-treated patients [184]. There were no differences in the progression of the EDSS between treatment groups [184].

3.3 | Fumarates

Eight studies focused on remyelination and neuroprotection by DMF. DMF was not observed to induce OPC differentiation under either basal or inflammatory conditions in mouse OPCs [166]. Nevertheless, DMF has demonstrated a protective effect against oxidative stress and nitric oxide bursts—mediated via *the Nrf*2 pat*hway*—in both in vitro and in vivo models [120, 122, 185, 186]. In an EAE model, DMF treatment improved clinical symptoms and preserved myelin [185]. Another in vivo study using an EAE model showed less demyelination when DMF was combined with IFNβ compared to monotherapy (either DMF or IFNβ) [168]. In the CPZ model, remyelination measured by staining for myelin proteins was slightly accelerated after treatment with DMF or monomethylfumarate (MMF) in the corpus callosum but not in the cortex. Neither DMF nor MMF affected OPC or ODC density, axonal injury, microgliosis, or astrocytosis [186]. Both diroximel fumarate (DRF) and DMF modulated neuronal network hyperexcitability caused by CPZ during remyelination [168, 185–188].

A single post hoc MTR analysis from a multicenter RCT investigating the efficacy and safety of DMF [189] revealed an increase in whole brain volume and NABT in the treated group, suggestive of remyelination, while MTR reductions were observed in the placebo group. This effect was also seen in patients without clinical or radiographic activity, suggesting it is independent of spontaneous lesion remyelination [190].

3.4 | Sphingosine-1-Phosphate Receptor Modulators

The sphingosine-1-phosphate receptor (S1PR) family consists of five receptors which are variably expressed in the immune system, the cardiovascular system, and the CNS [191, 192]. Table 4 shows the effects of S1PR modulators and their expression on different cell types [148, 193].

3.4.1 | Fingolimod (FTY)

Fingolimod (FTY) was extensively studied ($n = 25$) preclinically. In vitro studies demonstrated a protective effect of FTY against n-methyl-D-aspartate (NMDA) toxicity and oxidative stress in neurons and astrocytes [194, 195]. Moreover, various studies confirmed a neuroregenerative effect of FTY in a dose and

TABLE 4 | S1PR: Expression in various cell types and S1PR modulators, the cell types that express them and the DMT that modulate them.

Receptor	Expression and effect	DMT
S1P1R	**Lymphocytes**: prevention egression of lymphocytes from lymph nodes into the blood Neurons: migration NPCs, neuronal development, and neurite outgrowth **Astrocytes**: proliferation and activation Oligodendrocytes Endothelial cells: Increasing permeability, vascular leakage Microglia: enhanced microgliosis, reduction activation	FTY SPN (high affinity) OZN (high affinity) PON (high affinity)
S1P2R	Widely expressed in different organs: Inhibition of apoptosis, cellular proliferation, actin remodeling, development of the heart and the auditory and vestibular system	
S1P3R	Lymphocytes, macrophages, dendritic cells, neurons, astrocytes, microglia, oligodendrocytes, endothelial cells, smooth muscle cells	FTY
S1P4R	Lymphocytes, macrophages, dendritic cells	FTY
S1P5R	Neurons, astrocytes, microglia **Oligodendrocytes**: OPC survival, migration, differentiation **NK cells**: regulation egress into blood	FTY SPN (high affinity) OZN (high affinity)

Note: In bold: high level of expression of the respective receptor in these cells.

duration-dependent manner [124, 196], which could be in part mediated by increased levels of BDNF in the CNS [197–200]. Additionally, FTY increased expression of genes involved in neuronal activity and morphology, further supporting its potential neuroprotective effect [125, 197]. High FTY concentrations increased the proliferation and migration of NPCs as well as increased the differentiation of OPCs. Low concentrations resulted in an increase in NPC differentiation and induction of morphological changes of OPCs [126, 127, 196, 201, 202]. Chronic treatment improved neurogenesis and increased the proliferation and survival of NPCs. The response induced by FTY was thought to be attributed to direct effects on OPCs and indirect effects on astrocytes and neurons via the p38MAPK, ERK1/2, and CREB pathways [130].

Despite the encouraging results from in vitro studies, in vivo animal studies showed discrepancies [124, 202–204]. In the LPC model, both reduction of demyelination and increase of the myelination marker myelin basic protein (MBP) have been found [128, 129, 131]. However, one study failed to show an effect of FTY on remyelination [132]. In an EAE model, FTY reduced demyelinated damage and promoted OPC proliferation and differentiation. Prophylactic treatment improved clinical outcomes, maintained normal neurotransmission, and preserved myelin density as measured by DTI compared to untreated mice. Therapeutic treatment also prevented further worsening of symptoms [135, 205–207]. In a dose–response study, FTY promoted OPC proliferation and differentiation in an EAE animal model [134]. One study demonstrated a protective effect on axonal damage accumulation but could not establish enhanced remyelination [135]. Longitudinal brain volumetry in EAE mice demonstrated a superior effect of FTY compared to TF in maintaining cerebellum and striatum volume [135, 199]. Although FTY following CPZ showed positive outcomes in organotypic cerebellar slices [127, 208], later studies did not find evidence of increased remyelination when using immunostaining and TEM compared to the natural repair observed in untreated animals after CPZ withdrawal [132, 136–139]. In the TMEV model, FTY increased the proliferation of NPCs but did not influence NPC differentiation into ODCs nor change the clinical disease phenotype [133].

Seven clinical studies investigated the remyelinating or neuroprotective effects of FTY. One post hoc analysis of a phase III double-blind placebo-controlled trial in RMS patients revealed a treatment effect on six-month confirmed EDSS progression. However, most of this effect was attributed to the treatment effect on new or enlarging T2 lesions and relapses [209]. One small retrospective study determined that FTY improved disability in younger patients with milder disability and established this by the presence of remyelination and the absence of newly evolving demyelinating lesions on qMM [210]. Another retrospective study demonstrated that FTY improved functional electrophysiological measurements in RMS patients after one year of treatment [211]. Treatment effect measured by VEP was confirmed by an RCT in patients treated with FTY after a first episode of optic neuritis [172]. A prospective study showed a stabilizing effect of FTY on DTI and MTR measurement in normal appearing white matter (NAWM) and lesions [212]. In a longitudinal MTR study, both FTY and NTZ increased white matter lesion MTR [213]. However, FTY patients showed gradual increases after 2 years, while NTZ patients experienced declines after 6 months. The observed MTR effects in FTY patients after 6 months indicate that reparative processes, in addition to anti-inflammatory mechanisms, may be involved [213]. In a non-controlled prospective study, DTI analysis found no improvement in whole brain myelin integrity with FTY, but did show repair in specific myelin tracts linked to patients' clinical impairment [214].

3.4.2 | Siponimod (SPN)

Remyelinating and neuroprotective effects of Siponimod (SPN) were investigated in nine preclinical studies. Like FTY, SPN has been suggested to have neuroprotective properties, as indicated

by its ability to prevent neurotoxic glutamate levels [215], prevent altered neurotransmission [216], and enhance the propagation of electrical stimuli [217]. In an EAE model, prophylactic treatment reduced degeneration of the inner retinal layers measured by OCT and attenuated the severity of the clinical phenotype, while therapeutic treatment reduced demyelination and severity of phenotype, but did not change the number of myelinating ODC [141, 143]. Several studies in the Xenopus tadpole model demonstrated increased remyelination through immunolabeling and TEM in the optic nerve and brainstem [21, 143]. In the CPZ model, one study found that SPN reduced demyelination, acute axonal injury, and glial activation through the S1P5 receptor, independent of T and B-cell modulation [218]. Another study investigating ongoing CPZ damage demonstrated that treatment with SPN resulted in recovery of myelin-related protein expression levels and protection of maturing ODCs, but no induction of OPC proliferation was shown [144]. Imaging of CPZ mice showed lower signal intensity in the corpus callosum after SPN treatment versus controls, suggesting enhanced remyelination [143]. In ex vivo patch clamp recordings, extracellular application of SPN modulated channel activity in the thalamocortical system and counteracted the state of hyperexcitability induced by CPZ [188]. A separate study examined the effects of combining SPN and vitamin D after CPZ-induced damage and found that combination therapy was associated with improvements in behavioral testing, an increased myelin area percentage in histology analysis, and higher MBP expression levels. In contrast, monotherapy resulted in variable outcomes in behavioral testing, less evident improvement in myelination, and no change in MBP expression [145].

A single phase-III RCT in secondary progressive MS (SPMS) patients showed a significant reduction of the risk of three- and six-month confirmed disability progression in SPN-treated patients compared to placebo-treated patients [219]. A retrospective imaging analysis of this trial demonstrated a significant reduction in the progression of whole brain and gray matter atrophy. An increase in MTR was revealed in all brain tissues 2 years after treatment, most pronounced in NAWM. In newly formed lesions, SPN was associated with improved MTR recovery compared to placebo [220].

3.4.3 | Ozanimod (OZN)

A single study investigated ozanimod (OZN) in EAE mice, where treatment led to a significant reduction in clinical disease scores along with decreased inflammation and demyelination in the spinal cord as observed through histology. After CPZ-induced damage, OZN led to an attenuation of myelin loss and axonal protection, but myelin content assessed with immunostaining was not enhanced in the cortex, hippocampus, or corpus callosum [192]. No clinical studies dedicated to the remyelinating effects of OZN on remyelination were retrieved in the literature search.

3.4.4 | Ponesimod (PON)

One preclinical study reported positive effects of PON both in vitro and in vivo. PON more effectively promoted mouse OPC differentiation than SPN, FTY, or OZN, but did not affect myelination or OPC migration. In the same study, it was observed that PON treatment improved VEP latency compared to the vehicle and ameliorated working memory deficits in mice following CPZ-induced damage. Histological analysis revealed an increase in myelination and a decrease in *g* ratio. No clear dose–response effect was observed [151]. No clinical studies dedicated to the remyelinating effects of PON on remyelination were retrieved in the literature search.

3.5 | Monoclonal Antibodies

3.5.1 | Natalizumab (NTZ)

No preclinical studies investigating NTZ and remyelination were retrieved in the literature search.

Six clinical research articles were retrieved. One meta-analysis determined that patients with NTZ have some reduction in disability progression at 24 months [221]. A study examining NTZ use for periods exceeding 5 years observed no association between the duration of NTZ treatment and the progression of brain atrophy [222]. A comparative observational study of NTZ and IFNβ has found that patients treated with NTZ showed a reduced rate of brain atrophy and less decline on neuropsychological tests [223]. A prospective open-label study reported that patients treated with NTZ exhibited a greater volume undergoing MTR increase compared to those treated with IFNβ and HCs, which may indicate remyelination [174]. In a separate MTR analysis involving patients treated with NTZ and FTY, an increase in lesional MTR was observed after 6 months in the NTZ group; however, MTR either decreased or remained stable during subsequent follow-up periods [213]. In a prospective study, DTI analysis revealed an improvement in white matter damage severity in NTZ-treated patients compared to treatment with GA and IFNβ [224].

One clinical trial (NCT05418010) is assessing the effect of NTZ in facilitating remyelination in highly active MS patients utilizing MTR analysis and VEP measurement.

3.5.2 | Ocrelizumab (OCR)

No preclinical studies investigating the effects of OCR on remyelination were retrieved in the literature search.

Two clinical studies evaluated the effects of OCR on remyelination. One retrospective analysis of data from OPERA I/II and ORATORIO [225, 226] established less thalamic volume loss in OCR-treated patients, and this effect was more pronounced in patients who started treatment earlier. Moreover, an association could be found between baseline thalamic volumes and clinical outcomes in RMS patients [227]. Another study employing MWF analysis found either an increase or stabilization in all brain regions of NAWM and an increase in chronic lesions among patients treated with OCR, indicating higher levels of myelin [228].

3.5.3 | Ofatumumab (OFA)

No studies on remyelinating effects of OFA were found.

3.6 | Immune Reconstitution Therapies

3.6.1 | Cladribine (CLAD)

One in vitro study evaluated OPC differentiation under both inflammatory and basal conditions after CLAD administration, but could not reveal a significant effect of CLAD on immunocytochemistry or MBP transcript levels [166].

Post hoc analysis of a phase III RCT in RMS patients revealed a significant reduction in the percentage brain volume change and a lower risk of disability progression in treated patients [229]. One retrospective study investigated the effect of several DMTs (IFNβ, GA, TF, DMF, FTY, CLAD, and OCR) on VEP latency and found no treatment effect and no clear difference between DMTs [230].

Currently, an open-label phase IV single center clinical trial (NCT05902429) is ongoing to assess remyelination through qMM in patients with highly active MS.

3.6.2 | Alemtuzumab (ALEM)

In vitro research, performed on MBP-stimulated cultures of peripheral blood mononuclear cells derived from ALEM-treated patients demonstrated increased concentration of neurotrophic factors (BDNF, platelet-derived growth factor and ciliary neurotrophic factor). Media from these cultures enhanced OPC survival, maturation and myelination assessed with immunostaining in rat ODC culture [164]. In an EAE model, ALEM was found to attenuate clinical scores, increase myelin coverage via immunostaining of the spinal cord and upregulate myelin genes [165].

Five clinical research articles were retrieved. Evaluation of treatment effects on the visual system over a two-year study period demonstrated an improvement in VEP latencies [231]. A prospective study, following patients 5 years after treatment demonstrated a decrease in retinal ganglion cell layer, measured with OCT [232]. This study also demonstrated a reduction of brain parenchymal fraction and gray matter volume, however white matter volume remained stable and myelin content, measured with synthetic MRI, was increased in patients without pre-baseline disease activity [232]. Another study using DTI analysis showed that treated patients had stable corpus callosum myelin structure over 2 years [233]. MTR analyses reported a decrease of the MTR ratio and worsening of the periventricular MTR gradient in untreated patients. In contrast, the MTR ratio stabilized [234] and periventricular gradient reversed [235] in patients treated with ALEM. Magnetic resonance spectroscopy demonstrated that N-acetyl aspartate concentrations remained stable over a two-year period, suggesting sustained metabolic integrity of the neuronal tissue [233]. In the same study, MWF analysis demonstrated stability over 2 years in both NAWM and lesions, indicating evidence of repair [233].

4 | Discussion

Combination of DMT and remyelinating interventions should be the path forward to optimize outcomes in MS patients and reduce disability progression. Our review underscores the paucity of preclinical investigations of combination treatments, suggesting a significant knowledge gap.

Preclinical research does suggest remyelinating and neuroprotective properties of various DMT, evidenced by decreased apoptosis, reduction of oxidative stress, increased secretion of neurotrophic factors, reduction of demyelination, preservation of myelin, and stabilization of neuronal networks. However, robust evidence of authentic remyelination—whether demonstrated through manipulation of ODC lineage cells in cell cultures or confirmed by histological analysis in animal models—has only been documented in studies utilizing GA, TF, FTY, SPN, PON, and ALEM.

Although validated biomarkers for assessing remyelination in clinical trials are currently lacking, several emerging tools are being utilized to monitor myelin repair and evaluate functional recovery [27]. Therefore, this systematic review included studies with various biomarkers to assess remyelination and neuroprotection. Positive effects on visual outcomes were reported in clinical studies using FTY, ALEM, and TF. However, these findings are based on limited sample sizes. Several studies used MTR analysis and reported beneficial effects with DMF, FTY, SPN, NTZ, and ALEM. Other advanced imaging techniques in clinical studies, namely DTI and MWF, showed favorable effects of treatment with GA, OCR, and NTZ. Studies using multiple advanced MRI techniques (DTI, MR spectroscopy, MWF) in ALEM-treated patients pointed toward stabilization in myelin structure. Considering all clinical study data in this review, both FTY and ALEM showed the most promising results on various outcome assessments. However, it should be mentioned that little or no comparative studies have been performed with more recent treatments (such as CLAD, OCR, OFA, and OZN). Additionally, by excluding clinical trials without any information posted in the last 2 years, it is possible some negative trials were not retrieved in the literature search.

In vitro and in vivo studies mostly focused on the impact of DMT on ODCs and their precursors. However, the complex process of remyelination is an interplay between ODC, microglia, and astrocytes [37–40]. In this regard, emerging treatment options such as BTK-i, which can regulate microglial function [236], could have an effect on remyelination as well. Recent evidence from ex vivo (organotypic cerebellar slices) and in vivo (transgenic Xenopus tadpole) studies demonstrates the potential of BTK-i to enhance remyelination both in monotherapy as well as in combination with Clemastine [237, 238].

Few studies explored combination treatments. However, the complex processes underlying MS pathology highlight a need for future studies to investigate combination treatments in vitro and in vivo.

One major limitation that should be considered when comparing results across studies is the observed heterogeneity of the preclinical models and primary and secondary outcome measures in the various included studies in this review, making the data not well-suited for meta-analysis. This further illustrates the need for a common framework of outcome measures to assess remyelination and neuroprotection, as it would benefit researchers, industry, and regulatory authorities to study novel treatment approaches. In preclinical research, newer in vitro models, such

as fully developed brain organoids, may be promising tools to screen combinations of regenerative and immunomodulatory treatments [239]. Most frequently used surrogate markers in clinical trials investigating remyelinating agents are clinical improvement or worsening, VEP, and advanced imaging techniques such as MTR and DTI [27]. However, it remains unclear whether changes in these assessments are caused exclusively by enhanced remyelination, as other confounders such as age-related atrophy, anti-inflammatory effects, and endogenous remyelination should be considered. Combining different modalities such as electrophysiology and advanced imaging techniques could enhance specificity for myelin repair assessment [240].

Numerous questions remain in optimizing trial design, including the choice of study population (earlier versus later stages of MS disease course), treatment administration dose, duration, and frequency, trial duration, and outcome measures. As remyelination is heterogeneous across patients and age-dependent, future research on remyelinating agents could benefit from stratifying patients by remyelination potential [240, 241]. Furthermore, as it is difficult to differentiate the anti-inflammatory effect using currently available outcome measures, using a lag time after the recent start of a DMT could control for an anti-inflammatory effect. Ultimately, to advance the development of remyelinating treatments, future research should focus on advancing novel biomarkers to assess and monitor repair mechanisms, for example, with the use of existing and new fluid biomarkers [242], liquid biopsies [243] including extracellular vesicles, and exploring advanced imaging techniques or using deep learning methods to generate myelin maps from standard MRI sequences [244].

5 | Conclusions

This review aimed to summarize the remyelinating and neuroprotective properties of current DMT in order to develop the most advantageous combination therapy. Preclinical data suggest that various DMT have neuroprotective properties, and some DMT showed evidence of enhanced remyelination. However, due to a lack of appropriate outcome measures for remyelination in clinical trials, no clear effect on remyelination could be shown. Therefore, research efforts should be focused on exploring and validating biomarkers to assess and monitor repair mechanisms in MS.

Conflicts of Interest

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Data Availability Statement

Data supporting this study are included within the article and Supporting Information.

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References [51] to [229] are cited in Supporting Information.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** ene70397-sup-0001-DataS1.xlsx. **Appendix S1:** ene70397-sup-0002-AppendixS1.docx.